

Sunbeam - GenAI Hackathon

Subject: Applied Generative AI/LLM Application Development

Title: Sunbeam Lab Exam Evaluation Assistant

Maximum Marks: 40

Problem Statement

Develop an AI-powered Sunbeam Lab Exam Evaluation Assistant that helps human evaluators assess programming laboratory submissions.

*The application should evaluate a student's solution using a provided **marking scheme** and generate section-wise marks along with supporting evidence.*

The system acts only as an assistant. Final grading remains with the human evaluator.

Functional Requirements

Develop an AI assistant capable of evaluating a programming laboratory submission.

Your application should:

- *accept a student's submission containing one or more source code files (as zip file).*
 - *use the supplied question paper and marking scheme to perform an evidence-based evaluation.*
 - *generate section-wise marks and evaluator comments.*
 - *calculate the final score (out of max marks).*
 - *present the evaluation in a structured, human-readable form.*
 - *allow the evaluation report to be saved for future reference.*
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General Guidelines

The implementation may use any suitable architecture including but not limited to:

- *Prompt Engineering*
- *Lang Chain*
- *Lang Graph*
- *Agentic AI*
- *Multi-Agent Systems*
- *Workflow Graphs*
- *RAG*
- *Custom Pipelines*

Students are free to use any local LLM (preferred) or online LLMs.

No marks will be awarded or deducted based on

- *coding style*
- *project structure*
- *framework selection*
- *naming conventions*

Different but technically correct implementations are acceptable.

Input

The evaluator receives

- *lab exam paper*
 - *marking scheme*
 - *(optional) reference solution*
 - *student submission*
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Output

Generate a structured evaluation containing

- Student ID
- Section-wise marks
- Maximum marks
- Evidence
- Evaluator comments
- Total marks

Marking Scheme

Sr. No.	Evaluation Area	Max Marks	What Will Be Evaluated
1	Documentation (RFADE.md)	7	Architecture Mermaid Diagram, Implementation Plan (Max 5 bullet points), and Actual Execution Steps (Max 10 bullet points).
2	Input Handling	7	Ability to correctly accept, organize, and prepare input data for further processing.
3	Evaluation Configuration	7	Appropriate use of evaluation criteria and effective preparation of the evaluation process.
4	Core Evaluation Functionality	12	Correctness of evaluation, support for partial marks, consistency, and meaningful justification of awarded marks.

<i>Sr. No.</i>	<i>Evaluation Area</i>	<i>Max Marks</i>	<i>What Will Be Evaluated</i>
5	<i>Evaluation Results</i>	4	<i>Clear, structured, and understandable presentation of evaluation outcomes.</i>
6	<i>Report Generation</i>	3	<i>Ability to present and/or save the evaluation report in a usable format.</i>
	<i>Total</i>	40	
